Project Ideation and Problem Statement

Detailed project document · Prepared for project completion and review

Project reference information

Item	Known project detail
Project	Health360 Clinical & Healthcare Risk Analysis
Dataset size	5,000 patient/person records
Dataset fields	17 columns; exact field names and definitions must be copied from the source dataset's header row/data dictionary
Analytics tool	Tableau
Visual assets	9 worksheets/visualizations and a dashboard; a Tableau story for Patient Risk & Lifestyle Insights was also part of the project workflow
Repository	https://github.com/en25cs3010918-ctrl/Health360-Healthcare-Risk.git
Website	https://en25cs3010918-ctrl.github.io/Health360-Healthcare-Risk/
Scope caution	This is an educational data-visualization project. It is not a clinically validated diagnostic, prognostic, or treatment system.

Executive summary

Health360 Clinical & Healthcare Risk Analysis is an educational analytics project that organizes a healthcare-related dataset into interactive Tableau views. Its purpose is to help a viewer explore patterns in patient/person risk indicators and lifestyle-related variables, understand how measures vary across the available records, and communicate observations in a structured visual format. The project contains a dataset described as 5,000 records and 17 columns. The exact meaning of each field must be taken from the original dataset documentation rather than inferred from its name.

Problem statement

Healthcare datasets can be difficult to interpret when information is presented only as rows in a spreadsheet. A user may need to compare groups, inspect distributions, notice data-quality limitations, and trace how an observation was produced. Health360 addresses this presentation problem by organizing the data into a dashboard and story. The project should not claim that visualization alone improves clinical outcomes or that associations found in the data establish cause and effect.

Proposed solution

Prepare and validate the dataset; define each measure and dimension; construct a set of Tableau worksheets; assemble them into a dashboard; use a story to sequence selected insights; and publish the views through Tableau Public or the approved hosting environment. A project website provides context and navigation to the published visualizations. Each conclusion should be linked to a specific chart, filter state, and data subset.

Intended users

Potential audiences include students, instructors, data-analytics reviewers, and readers interested in learning how healthcare-related data can be explored visually. Any use by clinicians or patients must be framed as exploratory only unless clinical validation, governance, and appropriate approvals are independently established.

Scope and exclusions

Included: data inventory, data cleaning, exploratory analysis, visualization design, dashboard/story development, publication, testing, and documentation. Excluded unless separately supported: clinical diagnosis, personalized treatment advice, prediction of an individual patient's outcome, causal claims, or claims that the dataset represents all patients or a general population.

Problem statement — final form

How can a healthcare-related dataset of 5,000 records and 17 fields be cleaned, described, and presented through an interactive Tableau dashboard and story so that users can explore available risk and lifestyle indicators while understanding the limits of the data?

Deliverables

A documented dataset and data dictionary; reproducible cleaning log; Tableau workbook containing worksheets and dashboard; a narrative story; a website linking to the visualizations; a test record; a final report and demonstration materials.

Completion and review record

Prepared by: _____ Date: _____

Reviewed by: _____ Status: Draft / Reviewed / Approved

Document-control note: Update this file when the dataset, workbook, published views, or website changes. Record version/date and keep claims tied to evidence. This detailed document uses known project context and explicitly flags details that must be verified from the actual source files.